AR(1) $\quad y_t = c + \phi_1 y_{t-1} + \varepsilon_t$

↳ news (unpredictable) shocks e.g. financial crisis.

$E(\varepsilon_t)=0 \Rightarrow$ LIE $\Rightarrow \varepsilon_t \sim$ MDS.

↓

ε_t < positive / negative. $\Rightarrow$ $\boxed{\text{focus.}}$ (1)

$$\varepsilon_t = \sigma_t u_t$$

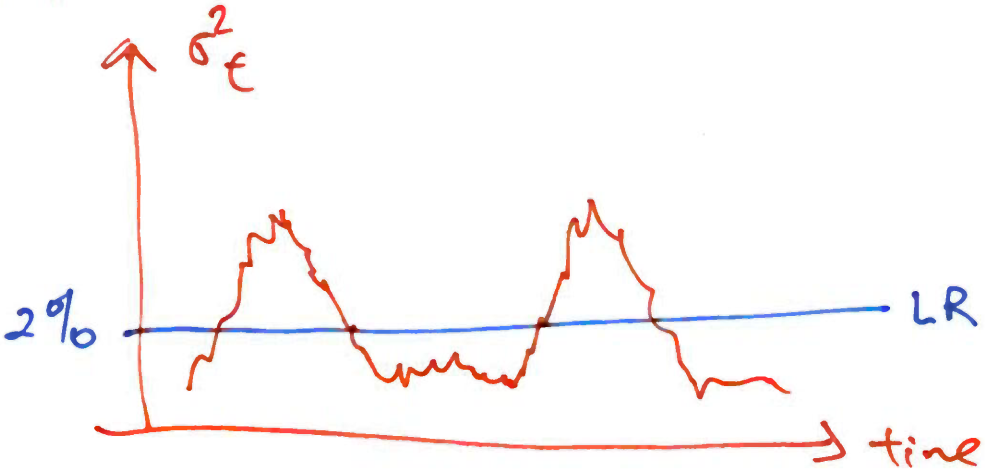
↳ week 8/9

SR variances $\Rightarrow$ $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \text{wow}\ \sigma_{t-1}^2$ GARCH(1,1)

wow = 0.9

$\boxed{\text{focus (2)}}$

↓



LR $\Rightarrow V(\varepsilon_t)$ derive now.

Assume $u_t \sim iid\ N(0,1)$

given $\boxed{\varepsilon_t \mid F_{t-1} \sim N(0, \sigma_t^2)}$

known given F_{t-1}

tells us that $\boxed{\sigma_t^2 = V(\varepsilon_t \mid F_{t-1}) = E(\varepsilon_t^2 \mid F_{t-1})}$

Derive

$$V(\varepsilon_t) = E(\varepsilon_t - \underbrace{E(\varepsilon_t)}_{0})^2 \qquad E(\varepsilon_t) = 0$$

$$= E(\varepsilon_t^2)$$

$$= E(E(\varepsilon_t^2 \mid F_{t-1})) \qquad \text{LIE}$$

①

$$= E(\sigma_t^2)$$

$$= E(\alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2)$$

$$= \alpha_0 + \alpha_1 E(\varepsilon_{t-1}^2) + \beta_1 E(\sigma_{t-1}^2)$$

② stationary.

$$= \alpha_0 + \alpha_1 E(\varepsilon_t^2) + \beta_1 E(\sigma_t^2)$$

$$= \alpha_0 + \alpha_1 V(\varepsilon_t) + \beta_1 V(\varepsilon_t)$$ ③

$$V(\varepsilon_t) = \frac{\alpha_0}{1-\alpha_1-\beta_1}$$ that's it.

Big Picture

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

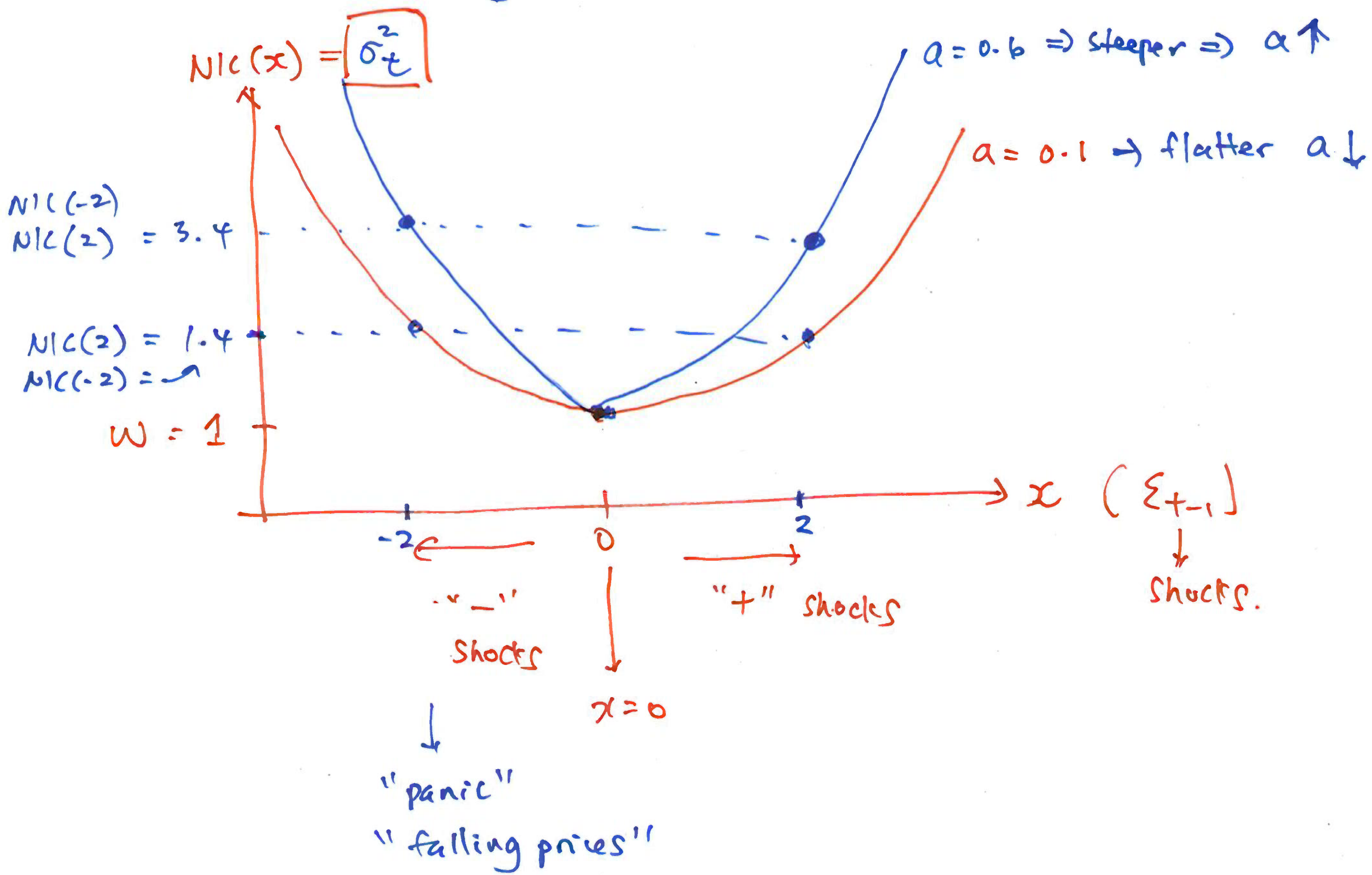
Trick!

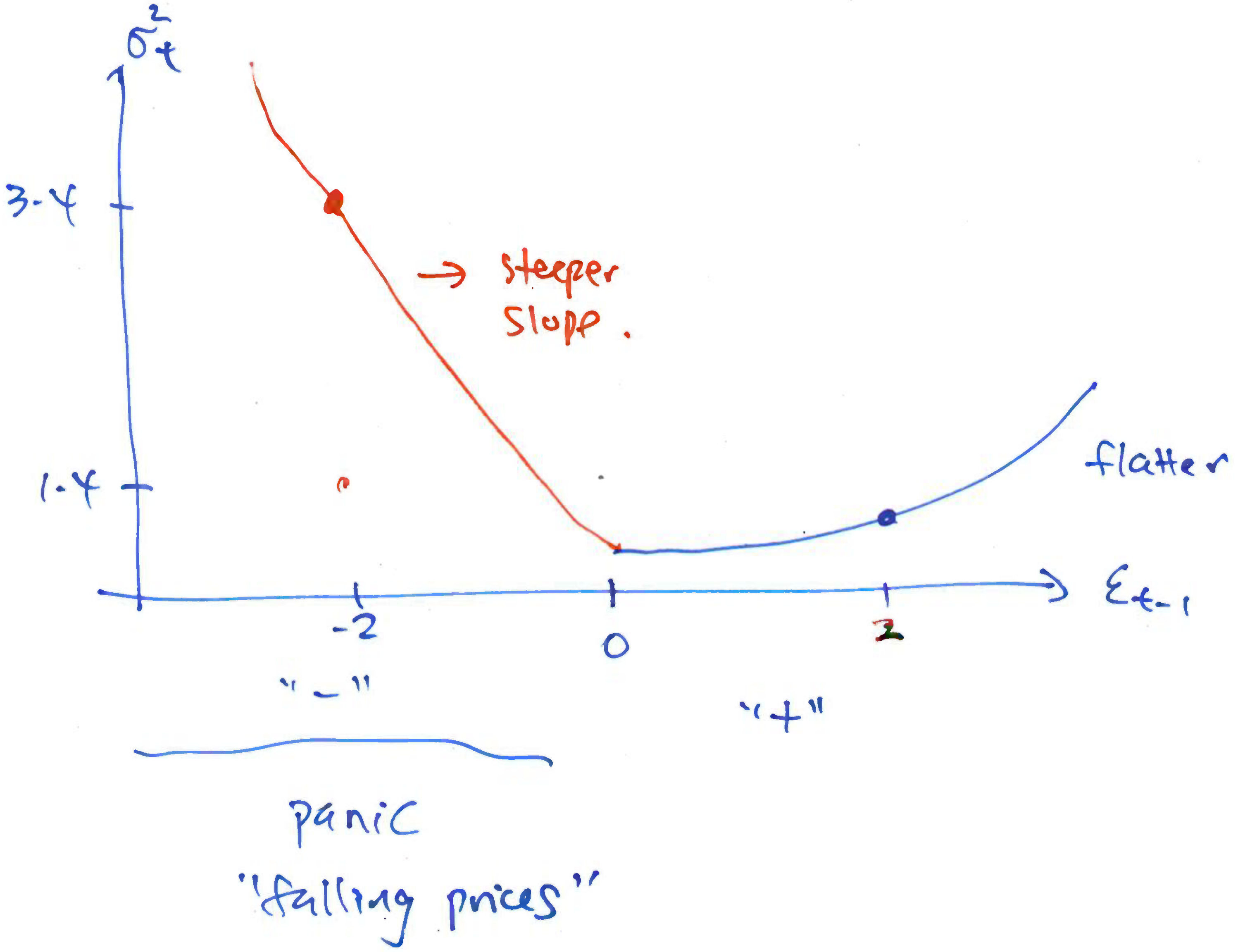
$$V(\varepsilon_t) = \alpha_0 + \alpha_1 V(\varepsilon_t) + \beta_1 V(\varepsilon_t) \Rightarrow$$ replace SR with LR.

$$V(\varepsilon_t) = \frac{\alpha_0}{1-\alpha_1-\beta_1} = 2\%.$$ -2%

$$\alpha_1 + \beta_1 < 1$$

$$NIC(x) = \underset{1}{\omega} + a\,x^2$$





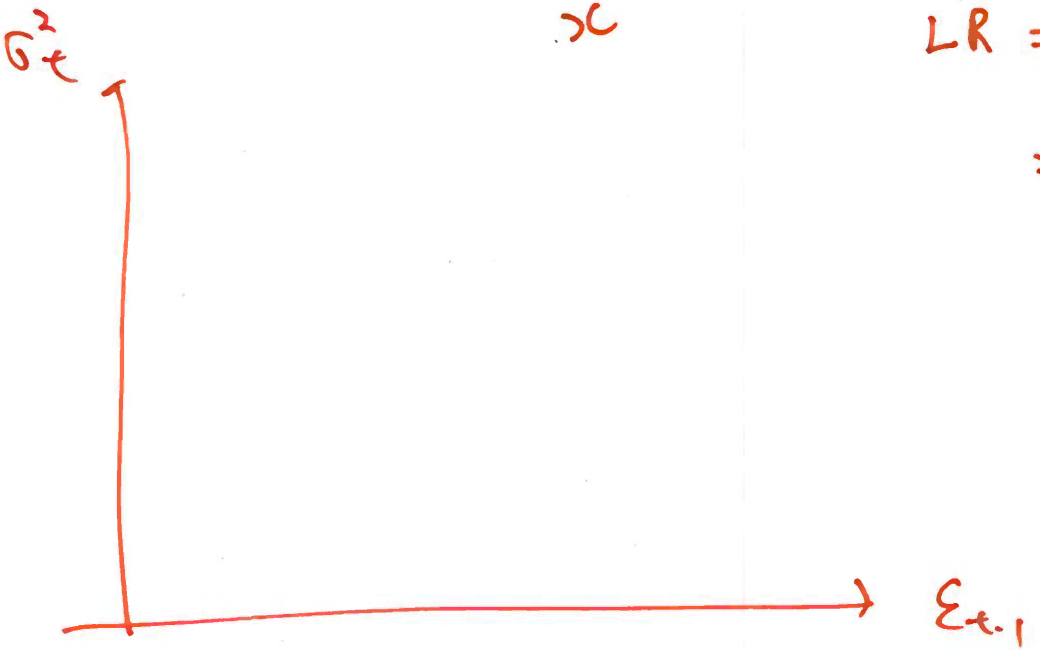
σ_t^2
3.4
1.4
→ steeper slope.
flatter
ε_{t-1}
-2
0
2
"-"
"+"
panic
"falling prices"

$$\frac{\partial^2 NIC(x)}{\partial x^2} = 2a$$

$$NIC(x) = \omega + a x^2 \qquad x = \varepsilon_{t-1}$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \underbrace{\varepsilon_{t-1}^2}_{x} + \beta_1 \underline{\underline{\sigma_{t-1}^2}}$$ → replace SR with LR

$$LR = V(\varepsilon_t) = \frac{\alpha_0}{1-\alpha_1-\beta_1} = 2\%$$



$$\sigma_t^2 = \underbrace{\alpha_0 + \beta_1\left(\frac{\alpha_0}{1-\alpha_1-\beta_1}\right)}_{\omega} + \alpha_1 \varepsilon_{t-1}^2$$

y

$x = \varepsilon_{t-1}$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \lambda \underbrace{I_{t-1} \varepsilon_{t-1}^2} + \text{wow } \sigma_{t-1}^2$$

known
given
F_{t-1}

$$I_{t-1} = \begin{cases} 1 & \text{if } \varepsilon_{t-1} \leq 0 \\ 0 & \text{if } \varepsilon_{t-1} > 0 \end{cases}$$

$$v(\varepsilon_t) = \alpha_0 + \alpha_1 v(\varepsilon_t) + \lambda \boxed{\tfrac{1}{2} v(\varepsilon_t)} + \beta_1 v(\varepsilon_t)$$

$$E\left(I_{t-1} \varepsilon_{t-1}^2\right) = \frac{1}{2} \underbrace{E\left(\varepsilon_{t-1}^2\right)}_{v(\varepsilon_t)}$$

$$v(\varepsilon_t) = \frac{\alpha_0}{1 - \alpha_1 - \lambda \frac{1}{2} + \beta_1}$$

LR $\Rightarrow$ GJR-GARCH.

$E(\varepsilon_t) = 0$

$v(\varepsilon_t) = \dfrac{\alpha_0}{1-\alpha_1-\beta_1}$

$u_t \sim iid\ N(0,1)$

very important

$$\boxed{\varepsilon_t \mid F_{t-1} \sim N(0, \sigma_t^2)}$$

① $v(\varepsilon_t \mid F_{t-1}) = E(\varepsilon_t^2 \mid F_{t-1}) = \sigma_t^2$

② $E(\varepsilon_t \mid F_{t-1}) = 0$ mds.

③ $E(\varepsilon_t^3 \mid F_{t-1}) = 0$

normal

$E(\varepsilon_t^3) = 0$

$E(\varepsilon_t^3) =$ stuck.

$= E\left(E(\varepsilon_t^3 \mid F_{t-1}) \right)$

$= E(0)$

$= 0$